

1 Interprocess communication

1.1 Producer-consumer buffer

Theorem 1.1. *If access to the buffer is protected by mutual exclusion, the number of elements remains between zero and the capacity N .*

$$0 \leq n_{\text{buffer}} \leq N \quad (1.1)$$



Figure 1.1. Data flow through the shared buffer

Table 1.1. Buffer semaphore state

Semaphore	Initial value
mutex	1
empty	N
full	0

```

1 def can_consume(items):
2     return items > 0
  
```

Listing 1.1. Checking buffer availability

Algorithm 1.1: Removing an element from the buffer

Input: A non-empty shared buffer

Output: The oldest element

```

1 wait(full)
2 wait(mutex)
3  $x \leftarrow$  remove element
4 signal(mutex)
5 signal(empty)
6 return  $x$ 
  
```

The invariant in [theorem 1.1](#), expressed by [equation \(1.1\)](#), prevents overflow and underflow. [Figure 1.1](#) summarizes the flow, while [table 1.1](#), [listing 1.1](#), and [algorithm 1.1](#) describe initialization and removal.